

GIVA: System Architecture Document

For Sustainable Circular Jewelry Platform

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Abstract

This document presents the comprehensive system architecture for GIVA, the leading full-stack circular jewelry platform focused on B2C sustainable jewelry, circular subscription, blockchain-verified recycled metals, and IoT-enabled circular take-back operations. The architecture is designed to support 2.5M+ active customers, 200+ experience stores, 500,000+ daily active users, 2.5M+ blockchain passports issued annually, \$1.28B+ in annual revenue, and 100,000+ circular take-back transactions per month by FY30F.

The architecture follows a microservices-based, cloud-native design with IoT-first data ingestion from circular take-back hubs, real-time blockchain passport issuance, and AI-driven personalization and inventory optimization. Key components include: (1) multi-channel customer onboarding system (web, mobile, in-store kiosk), (2) Circular Passport Engine for blockchain-based product verification, (3) CaaS (Circular-as-a-Service) billing and subscription management, (4) IoT-based circular take-back monitoring platform with RFID tracking, (5) Carbon credit bundling and offsetting engine, (6) AI-powered design recommendation and styling engine, (7) B2B partner portal for corporate gifting and Scope 3 reporting, (8) logistics and reverse logistics optimization engine, (9) inventory and demand forecasting platform, and (10) analytics and ESG reporting layer.

The architecture emphasizes scalability (horizontal scaling to 10M+ daily API requests), fault tolerance (99.95% uptime target, 2-hour RTO), security (AES-256 encryption, RBAC, PCI-DSS compliance for payments), and compliance with BIS hallmarking requirements, India CCTS, EU Digital Product Passport (DPP), and emerging circular economy regulations.

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1 Architecture Overview

GIVA’s system architecture is built on a microservices-based, cloud-native design deployed on AWS with edge processing for real-time circular take-back operations and inventory management. The architecture is designed to handle the unique challenges of circular jewelry: real-time product traceability via blockchain, circular take-back logistics, carbon credit integration, subscription billing, and AI-powered personalization across web, mobile, and in-store channels.

1.1 Design Principles

The architecture follows seven core design principles:

1. **Microservices-first:** Loosely coupled services that can be developed, deployed, and scaled independently. This allows GIVA to iterate quickly on the circular subscription engine without affecting product catalog or vice versa.
2. **Cloud-native:** Leveraging AWS managed services (EKS, RDS, S3, MSK, Lambda) for reduced operational overhead. This reduces the need for in-house infrastructure management and allows the engineering team to focus on domain-specific challenges.
3. **IoT-first:** Real-time ingestion and processing of telemetry from circular take-back hubs, RFID-tagged products, and in-store sensors. Every returned product generates 100+ data points—product condition, refurbishment status, carbon savings, circular journey.
4. **Blockchain-first:** Every product gets an immutable Circular Passport on Hyperledger Fabric, providing tamper-proof provenance from recycled source to end customer to circular return. This is the foundation of GIVA’s trust and premium pricing.
5. **AI-first:** ML models for design recommendation (collaborative filtering), demand forecasting (time-series), customer churn prediction, and carbon footprint calculation. These models improve over time as more data is collected, creating a defensible data moat.
6. **Security-by-design:** Encryption at rest and in transit, RBAC, comprehensive audit logging for financial transactions (payments, circular credits, carbon offsets). PCI-DSS compliance for payment processing.
7. **API-first:** Well-documented RESTful APIs for all external integrations. This enables corporate clients to integrate GIVA data into their own ESG reporting systems and allows carbon credit buyers to programmatically purchase offsets.

1.2 User Personas Supported

The architecture supports seven primary user personas, each with distinct capabilities and access patterns:

Table 1.1: User Personas and Capabilities

Persona	Type	Key Capabilities
Retail Customers	B2C	Browse catalog, purchase recycled jewelry, track circular passport, initiate returns, offset carbon, manage subscription
Corporate Clients	B2B	Bulk gifting, Scope 3 reporting, carbon credit bundling, employee subscription management
Store Operations	Internal	Inventory management, circular take-back processing, customer service, store analytics
Field Service	Internal	Returns pickup, refurbishment tracking, quality inspection, parts ordering
Circular Ops	Internal	Take-back monitoring, refurbishment workflow, scrap metal reconciliation, blockchain passport updates
Data API Users	External (corporate)	Aggregated circular economy data, market insights, ESG reports
Registry Auditors	Verra, Gold Standard, BIS	Circular verification, credit audit, MRV validation, hallmark compliance

1.3 High-Level Architecture Layers

The architecture comprises six layers, each with specific responsibilities:

Table 1.2: Architecture Layers

Layer	Components
Client Layer	Web app (React.js), mobile app (React Native iOS/Android), in-store kiosk (React), corporate API access (REST + GraphQL), registry auditor portal (read-only)
API Gateway Layer	Kong / AWS API Gateway for routing, rate limiting (100-10k req/min), authentication (JWT, API keys), observability (request logging, tracing)
Microservices Layer	12+ services including Product Catalog, Circular Passport Engine, Subscription Billing, Carbon Credit Engine, Take-Back Management, Inventory Optimization, AI Styling, Logistics, Analytics
Data Layer	PostgreSQL (operational - customers, orders, inventory), TimescaleDB (time-series for circular take-back), Redis (caching - sessions, product catalog), Snowflake (warehouse - analytics), S3 (data lake - product images, KYC), Hyperledger Fabric (blockchain - Circular Passport audit trail)
External Integrations	Payment gateways (Stripe, Razorpay, UPI), carbon registries (Verra, Gold Standard, CCTS), logistics providers (Delhivery, Shiprocket, Blue Dart), ESG platforms (Salesforce Net Zero Cloud, Persefoni), map APIs (Google Maps, Ola Maps)
Deployment & Observability	EKS on AWS (multi-region: ap-south-1, us-east-1, eu-west-2), CI/CD (GitHub Actions + ArgoCD), monitoring (Datadog, Prometheus, ELK, PagerDuty)

2 Client Layer

2.1 Customer Web Portal (Web)

Technology: React.js with TypeScript, Tailwind CSS, Next.js for SSR/SEO, WebSocket (Socket.io) for real-time order tracking.

Features:

- **Product catalog:** AI-powered personalized recommendations, real-time inventory availability, 360° product viewer, AR try-on (WebXR)
- **Circular passport:** Blockchain-verified product provenance: recycled source (refiner ID), carbon saved (vs. virgin mining), circular journey (if pre-owned), blockchain transaction hash
- **Carbon offset at checkout:** Real-time carbon footprint calculation, one-click offset purchase via Gold Standard credits, monthly impact report
- **Circular subscription management:** "GIVA Circle" subscription (999/month), unlimited jewelry rotation (3 pieces at a time), schedule swaps, track shipments
- **Circular take-back:** Initiate return for circular credit (70% of original price), print shipping label (prepaid), track return status
- **Order tracking:** Real-time order status (confirmed → manufacturing → shipped → delivered → return window), push notifications via WebSocket
- **Customer dashboard:** Purchase history, circular credit balance, carbon offset history, subscription status, circular passport collection

2.2 Mobile App (iOS/Android)

Technology: React Native + TypeScript + Redux Toolkit + offline-first SQLite (WatermelonDB) for local persistence.

Features:

- **Browse & purchase:** Same catalog as web, optimized for mobile (gestures, haptics), Apple Pay/Google Pay integration
- **AR try-on:** Real-time virtual try-on (Apple ARKit / Google ARCore) for rings, earrings, pendants. 95% accuracy in size estimation.
- **Circular passport scan:** Scan QR code on product packaging → instantly view full circular journey. Works offline (cached on device).
- **Subscription management:** Pause/resume GIVA Circle subscription, swap items (select next 3 pieces from catalog), schedule delivery dates

- **Circular return:** Initiate return by scanning product QR code, print shipping label (via email or QR at courier), track return status with push notifications
- **Offline mode:** Browse cached catalog (last 500 products), add to cart (sync when online), view cached circular passports (last 50 scanned)
- **Referral program:** Share referral code via WhatsApp/Instagram, track referral status (pending/completed), redeem circular credits

2.3 In-Store Kiosk

Technology: React.js (touch-optimized) + Electron (for POS integration) + Bluetooth (for RFID scanner).

Features:

- **Product discovery:** Browse catalog (large touch interface), filter by metal type (silver/gold/lab-diamond), price range, style (minimalist/bohemian/traditional)
- **Circular passport printing:** Customer scans QR code → prints physical Circular Passport (with QR code for digital verification)
- **Circular take-back processing:** Scan returned product RFID tag → validate authenticity (blockchain lookup) → assess condition (store associate input) → instant circular credit to customer account
- **Subscription signup:** In-store signup for GIVA Circle (999/month), associate-assisted onboarding, instant welcome gift (500 credit)
- **Inventory lookup:** Real-time inventory at this store (for immediate pickup), other stores (for transfer), central warehouse (for delivery)
- **POS integration:** Checkout (credit/debit card, UPI, circular credit), print receipt, email receipt, update inventory (real-time)

2.4 Corporate API Access

Technology: RESTful APIs (OpenAPI 3.0 specification) + GraphQL (for complex portfolio queries) + API keys + OAuth2 (client credentials).

Authentication: API keys (rotated every 90 days) with scope-limited permissions (read/write/admin). Corporate clients can generate multiple keys for different teams (procurement, ESG reporting, finance).

Use cases:

- **Corporate gifting:** Bulk order placement (POST /v2/corporate/orders), employee rosters (POST /v2/corporate/employees), gift tracking (GET /v2/corporate/orders/id/status)
- **ESG data aggregators:** Automated Scope 3 emissions data ingestion (GET /v2/esg/carbon-footprint, /v2/esg/recycled-content) for inclusion in broader ESG datasets (Bloomberg, MSCI, Sustainalytics)
- **Carbon credit buyers:** Programmatic carbon offset purchase (POST /v2/carbon/offset), retirement management (POST /v2/carbon/retire), portfolio performance (GET /v2/carbon/portfolio)
- **Research institutions:** Anonymized circular economy data access (supply-demand forecasts, return rates, carbon savings) under academic license (free, with attribution)

- **Government regulators:** Compliance reporting (GET /v2/compliance/bis, /v2/compliance/ccts), audit trail access (GET /v2/audit/transactions) with read-only permissions

3 API Gateway Layer

The API Gateway serves as the single entry point for all client requests, providing authentication, rate limiting, request validation, and observability.

Table 3.1: API Gateway Configuration

Component	Specification
Technology	Kong / AWS API Gateway
Authentication	JWT (customers, ops); API keys (corporate); OAuth2 (B2B); mTLS (internal, auditors)
Rate Limiting	100/min (customers); 1k/min (ops); 5k/min (corporate API); 50k/min (internal)
SSL/TLS	TLS 1.3 only; auto renewal via ACM (90-day)
DDoS Protection	AWS Shield Advanced + WAF (OWASP Top 10, IP reputation, SQLi, rate-based rules)
Request Validation	JSON schema (draft-07); limits: 1MB (customer), 10MB (corporate)
Logging	ELK stack; 90d hot storage, 1y cold (Glacier)
Caching	Redis cluster (3 shards, 2 replicas) for product catalog, sessions, rate limits

3.1 API Versioning Strategy

- **URI versioning:** /v1/products, /v2/products. Major version changes indicate breaking changes (schema changes, removed endpoints). Minor version changes (bug fixes, performance) do not change URI.
- **Deprecation policy:** Announce 6 months in advance via email to all impacted API consumers; maintain backward compatibility for 12 months after deprecation announcement.
- **Current versions:** v1 (stable, 99.95% uptime over last 12 months), v2 (beta for new features including GraphQL support, WebSocket streaming for order tracking)

4 Microservices Layer

4.0.1 Microservices Communication Architecture

The microservices layer comprises loosely coupled, domain-driven services that can be scaled independently based on specific load patterns, traffic volumes, and resource requirements. Unlike monolithic architectures where a single component failure can cascade across the entire system, GIVA’s microservices architecture ensures fault isolation—if the recommendation engine experiences latency spikes, the checkout and payment services remain completely unaffected. Services communicate via two primary mechanisms: gRPC for synchronous, high-performance internal service-to-service calls requiring immediate responses, and Apache Kafka for asynchronous, durable, event-driven workflows that demand reliability, replayability, and decoupling between producers and consumers.

gRPC (Synchronous Communication): Used for request-response patterns where low latency and strong consistency are critical. Examples include: product catalog lookup during checkout, circular credit balance verification before subscription billing, and inventory availability checks during order placement. gRPC leverages Protocol Buffers (protobuf) for efficient serialization (up to 10x faster than JSON) and HTTP/2 for multiplexed connections, reducing network overhead.

Apache Kafka (Asynchronous Communication): Used for event-driven workflows where services need to react to state changes without blocking the originating service. Examples include: when a customer places an order, an "order.placed" event is published to Kafka—the inventory service consumes this to decrement stock, the analytics service to update business KPIs, the notification service to send confirmation emails, and the circular passport service to update blockchain records. Kafka provides durability (messages persisted to disk), replayability (consumers can re-read from any offset), and decoupling (producers don’t need to know about consumers).

4.1 Microservices Summary

Table 4.1: Microservices Overview

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Service	Stack	Priority	Function
Product Catalog	Node.js + Elasticsearch + Redis	H	Product metadata, inventory, search, filtering, personalization
Circular Passport Engine	Go + Hyperledger Fabric + IPFS	H	Blockchain passport issuance (2.5M+/year), QR code generation, provenance verification
Subscription Billing	Node.js + Stripe + PostgreSQL	H	GIVA Circle billing (999/month), recurring payments, dunning management, revenue recognition
Carbon Credit Engine	Python FastAPI + Gold Standard API	H	Real-time carbon footprint calculation, credit purchase, offset retirement, impact reporting
Take-Back Management	Python + PostgreSQL + Redis	H	Return initiation, shipping label generation, circular credit calculation, refurbishment workflow
Inventory Optimization	Python + Prophet + XGBoost	H	Demand forecasting (SKU-level, store-level), safety stock, auto-reorder, warehouse allocation
AI Styling Engine	Python + TensorFlow + Feast	M	Design recommendation (collaborative filtering), personalized product ranking, outfit completion
Logistics	Go + OR-Tools + PostGIS	M	Route optimization (pickup/delivery), courier allocation, real-time tracking, ETA prediction
Analytics & Reporting	Python + dbt + Superset	M	BI dashboards, ESG reports (Scope 3), executive KPIs, circular economy metrics
Notification Service	Node.js + Bull + Resend	L	Multi-channel alerts (email, SMS, push, WhatsApp)
Fraud Detection	Python + XGBoost + Seldon	M	Return fraud detection (AI image recognition), payment fraud, coupon abuse
Blockchain Verifier	Go + Hyperledger Fabric	H	Circular Passport audit trail, tamper-proof MRV, smart contracts for circular credit distribution

4.2 Service 1: Product Catalog Service

Handles product metadata, inventory, search, and personalization. This is the most frequently accessed service (10M+ daily requests by FY30F).

Technology: Node.js 20.x + Elasticsearch 8.x (search) + Redis Enterprise (caching) + PostgreSQL (source of truth).

Key functions:

- **Product metadata management:** CRUD for product catalog (SKUs: 10,000+ by FY30F). Fields: name, description, metal type (silver/gold/platinum/lab-diamond), weight (grams), dimensions, images (10+ per SKU, 4K resolution), videos (3D model), price (INR/USD), circular credit value, carbon footprint.
- **Inventory management:** Real-time stock by location (warehouse: Mumbai, Jaipur; 80+ stores). Inventory updates via Kafka (topic: inventory.changes). Safety stock calculation (95% service level, 7-day lead time).
- **Search & filtering:** Elasticsearch for full-text search (product name, description, tags), faceted filters (price range, metal type, style, occasion), geo-search (find in nearby stores). Target latency: <200ms (p95).
- **Personalization:** AI-powered product ranking based on customer history (clicks, purchases, returns, circular credit usage). Model: Two-tower neural network (user tower, item tower). Real-time inference via TensorFlow Serving.

4.3 Service 2: Circular Passport Engine

Handles blockchain-based product verification and provenance. This is the core of GIVA's trust and premium pricing.

Technology: Go 1.21 + Hyperledger Fabric 2.5 (blockchain) + IPFS (decentralized image storage) + PostgreSQL.

Key functions:

- **Passport issuance:** Every product gets an immutable Circular Passport at manufacturing. Data stored on blockchain: unique product ID (UUID), recycled source (refiner ID, batch number, certificate hash), carbon saved (tCO₂e vs. virgin baseline), manufacturing date, initial circular credit value. IPFS hash for product images (decentralized storage).
- **QR code generation:** Each passport has a unique QR code (printed on product tag/packaging). QR encodes: blockchain transaction ID (txn_{id}), *productID*, *verificationURL* (<https://passport.giva.com>).
- **Verification:** Customer scans QR code → API calls blockchain node → returns passport data in <500ms. Verification endpoint rate: 500,000+ scans/month (FY26E), 5M+ by FY30F.
- **Circular journey updates:** When product is returned via take-back, new transaction added to blockchain (return date, refurbishment details, new circular credit value, resale date). Immutable history append-only.
- **Smart contracts for circular credit:** When product is returned, smart contract automatically calculates circular credit (70% of original purchase price) and distributes to customer's wallet. Executes on blockchain, eliminating reconciliation.

4.4 Service 3: Subscription Billing Service

Handles GIVA Circle subscription management and recurring billing.

Technology: Node.js 20.x + Stripe Connect (payment processing) + PostgreSQL + Redis (session management).

Key functions:

- **Subscription lifecycle:** Signup (customer selects plan: monthly 999, annual 9,990 + 2 months free), pause (up to 3 months per year, no charge), resume, cancel (immediate or end-of-billing-cycle)
- **Recurring billing:** Stripe integration for card payments (Visa, Mastercard, Amex, RuPay) and UPI mandates (automatic debit). Retry logic: 3 retries (3 days, 7 days, 14 days). Dunning management: email reminders at 3, 7, 14 days past due, then SMS at 21 days, then service suspension at 30 days.
- **Usage tracking:** Track number of items out with customer (max 3 at a time), swap frequency (unlimited within limit), keep/buyout events (after 12 months, 50% of retail).
- **Revenue recognition:** ASC 606 compliant. Monthly fee recognized ratably over month. Keep/buyout revenue recognized at point of sale.

4.5 Service 4: Carbon Credit Engine

Handles real-time carbon footprint calculation and carbon offset purchases.

Technology: Python FastAPI + Gold Standard API + Verra API + Redis (price caching).

Key functions:

- **Real-time carbon footprint calculation:** At checkout, calculate: (product weight in grams $\times$ recycled metal emission factor (GIVA's verified factor: 2.5kg CO per \$100 of recycled jewelry vs. 25kg for virgin) $\times$ shipping distance (warehouse to customer). Latency: ≤ 200 ms.
- **Credit sourcing:** Bulk purchase of Gold Standard credits (50,000 tonnes/year at \$8-12/tonne) via API. Credits stored in GIVA's registry account, retired on customer's behalf at checkout.
- **Consumer offsetting:** "Make this purchase carbon-neutral for XX" — 15% conversion rate, \$2-5 incremental AOV. 100% margin (cost passed to consumer).
- **Retirement certificates:** Automated certificate generation (PDF + JSON-LD) with full provenance: customer name, order ID, tonnes offset, project details (Gold Standard project ID), retirement transaction hash. Email to customer within 5 minutes.
- **Corporate offtake:** Manage 5-10 year offtake agreements with corporate buyers (Microsoft, Stripe, JPMorgan). Track delivery against annual quotas, generate offtake certificates, automate invoicing.

4.6 Service 5: Take-Back Management Service

Handles circular take-back initiation, processing, and refurbishment workflow.

Technology: Python + PostgreSQL + Redis + Celery (async tasks) + Shiprocket API (shipping labels).

Key functions:

- **Return initiation:** Customer initiates return via web/mobile/kiosk → select product (from purchase history) → reason code (size issue, design mismatch, damaged, no longer wanted) → upload photo (optional) → AI fraud detection (image comparison).
- **Circular credit calculation:** Smart contract calculates 70% of original purchase price (in circular credits). Credit expires in 12 months if unused. Credit can be applied to any future purchase (excluding subscription fees).
- **Shipping label generation:** Integrate with Shiprocket/Delhivery API → generate pre-paid return label (email to customer, downloadable PDF). Free pickup from customer address (pin code validation).
- **Refurbishment workflow:** Return received at refurbishment center (Bangalore/Mumbai) → RFID scan → automated grading via AI image recognition (10+ images, 98% accuracy) → condition assessment (Excellent/Good/Fair/Poor) → refurbishment steps (cleaning, polishing, stone tightening, plating) → quality inspection → re-listing on "GIVA Certified Pre-Owned" section.
- **Blockchain update:** After refurbishment, new transaction added to Circular Passport: return date, refurbishment details, new condition grade, resale price.
- **Scrap reconciliation:** Unsellable pieces (30% of returns) → sent to refiner (MMTC-PAMP) → scrap credit tracked via API → reconciled against circular credit liability.

4.7 Service 6: Inventory Optimization Service

Predicts demand and optimizes inventory across warehouse and 80+ stores.

Technology: Python + Prophet (Facebook time-series) + XGBoost + PostgreSQL + Grafana.

Key functions:

- **Demand forecasting:** SKU-level demand forecast for next 90 days (daily resolution). Model features: historical sales (24 months), seasonality (festivals: Diwali, Dhanteras, Raksha Bandhan, Valentine's, Mother's Day), promotions, competitor pricing, macro-economic indicators (GDP growth, consumer sentiment). Accuracy: 85% MAPE at 30 days, 75% at 90 days.
- **Safety stock calculation:** Service level target: 95% (i.e., 95% of demand met from stock). Lead time: 7 days (warehouse to store). Safety stock formula: $z\text{-score} \times \text{standard deviation of demand} \times \sqrt{\text{lead time}}$. $z\text{-score for 95\% service level} = 1.645$.
- **Auto-reorder:** When inventory falls below reorder point, automatically generate purchase order to manufacturing facility (Jaipur/Mumbai/Bangkok). Reorder point = (average daily demand $\times$ lead time) + safety stock.
- **Warehouse allocation:** Optimize allocation of new inventory to 3 warehouses (Mumbai — West, Jaipur — North, Bangalore — South) based on historical demand by region. Target: minimize inter-warehouse transfers (cost: 15/kg).
- **Store replenishment:** Weekly replenishment from regional warehouse to 80+ stores. Route optimization for delivery trucks (minimize distance, maximize fill rate). Real-time inventory sync via Kafka.

4.8 Service 7: AI Styling Engine

Provides personalized product recommendations and styling suggestions.

Technology: Python + TensorFlow 2.x + Feast (feature store) + Redis (real-time inference cache).

Key functions:

- **Personalized recommendations:** Two-tower neural network (user tower: 50+ features including past purchases, browsing history, return behavior, circular credit usage; item tower: 30+ features including metal type, style, price, popularity). Real-time inference latency: ≤ 100 ms (p95).
- **Outfit completion:** "Complete the look" — recommend earrings to go with a necklace, ring to go with a bracelet. Uses association rule mining (Apriori algorithm) on purchase co-occurrence data.
- **Style quiz:** Customer onboarding quiz (15 questions) to capture style preferences (minimalist/bohemian/traditional/contemporary), metal preference (silver/gold/rose gold), occasion frequency (daily wear/festival/wedding/gifting). Embedding model maps quiz responses to product space.
- **Seasonal trend detection:** Real-time analysis of social media (Instagram, Pinterest) and search trends (Google Trends) to identify emerging styles. Signal fed to design team (new collection planning) and inventory team (demand forecast adjustment).

4.9 Service 8: Logistics Service

Manages forward logistics (delivery) and reverse logistics (returns pickups).

Technology: Go 1.21 + Google OR-Tools (vehicle routing problem solver) + PostGIS (spatial) + Redis (real-time tracking).

Key functions:

- **Courier allocation:** For each order, select optimal courier partner (Delhivery, Shiprocket, Blue Dart, DTDC, India Post) based on: cost (/kg), delivery time (days), success rate (
- **Route optimization:** For store replenishment, solve vehicle routing problem (minimize total distance, maximize fill rate, respect delivery time windows). 80+ stores, 3 warehouses. Solver: OR-Tools (VRPTW). Solution time: ≤ 30 seconds for full batch.
- **Real-time tracking:** Integrate with courier APIs for real-time tracking updates. Customer-facing map with live location (when available). ETA predictions based on historical delivery data (ML model).
- **Return pickup scheduling:** For circular take-back returns, schedule free pickup from customer address. Time windows: 9AM-1PM, 1PM-5PM, 5PM-9PM. Assign to courier based on pin code coverage. SMS/WhatsApp confirmation with 1-hour pickup window.

5 Data Layer

The data layer consists of multiple storage systems optimized for different use cases (polyglot persistence), each chosen for specific access patterns and consistency requirements.

5.1 Operational Databases

Table 5.1: Operational Databases

Database	Technology	Purpose	Size (FY30F)
Primary OLTP	PostgreSQL 15	Customers, orders, inventory, circular credits, subscriptions, payments (ACID compliant)	15 TB
Time-series	TimescaleDB (PostgreSQL ext.)	Circular take-back telemetry: return rates, refurbishment times, scrap yields (5M+ readings/day)	10 TB
Blockchain	Hyperledger Fabric 2.5	Circular Passport audit trail, immutable records, append-only, cryptographically linked	25 TB
Caching	Redis Enterprise (cluster)	Sessions, product catalog cache, rate limits, real-time inventory (1ms latency)	250 GB
Search	Elasticsearch 8.x	Product search, customer search, audit logs, full-text search on support tickets	3 TB

5.2 Data Warehouse

Technology: Snowflake (primary) with auto-scaling (X-Small to 3X-Large warehouses). Amazon Redshift (legacy, being migrated).

Data sources: All microservices stream data to Kafka (30+ topics) → transformed via dbt (80+ models, SQL-based) → loaded to Snowflake.

Data volume: 5 TB raw data per month (FY26E) → 50 TB by FY30F (10x growth).

Refresh frequency: Real-time (streaming via Snowpipe) for operational dashboards; daily for analytical reports (3 AM UTC, 1 hour batch window).

Schema: Star schema with fact tables (*sales_{fact}*, *return_{fact}*, *subscription_{fact}*, *carbon_{offset}_{fact}*) and dimensions.

5.3 Data Lake

Technology: Amazon S3 (intelligent-tiering) + Glacier Deep Archive for cold storage.

Content:

- Product images (4K resolution): Retained indefinitely (current: 5 TB, projected: 50 TB)
- Customer KYC documents: Encrypted at rest (AES-256), retained 7 years post-last-transaction (RBI NBFC requirement)
- Circular Passport audit trail: Retained indefinitely (Verra/Gold Standard requirement, blockchain)
- System logs (application, access, audit): Retained 30 days in S3 Standard, 90 days in S3 Glacier Instant Retrieval, 1 year in Glacier Deep Archive
- ML model artifacts: Versioned (model registry), retained indefinitely (for reproducibility)

6 Scalability & Performance

6.1 Scaling Strategy

The architecture is designed to scale horizontally to handle 5x growth by FY30F. Each microservice can be scaled independently based on its specific load patterns.

Table 6.1: Horizontal Scaling Targets by FY30F

Component	FY26E	FY30F
Active customers (monthly)	850,000	2.5M
Daily active users	250,000	1,000,000
Daily API requests	1,000,000	10,000,000
Daily orders	15,000	50,000
Circular take-back returns (monthly)	30,000	100,000
Circular Passport scans (monthly)	500,000	5,000,000
Data warehouse size (TB)	10	50
Concurrent WebSocket connections	5,000	25,000

6.2 Performance SLAs

Table 6.2: Performance Service Level Agreements (SLAs)

Service	Target (p95)	Current (p95)
Product catalog search	≤200 ms	150 ms
Checkout (incl. carbon calculation)	≤1 sec	600 ms
Circular Passport verification (QR scan)	≤500 ms	300 ms
Order placement	≤2 sec	1.2 sec
Circular credit issuance (return)	≤5 sec	3 sec
Subscription billing (monthly)	≤10 sec	5 sec
AI recommendation (personalized)	≤100 ms	70 ms
Inventory lookup (real-time)	≤50 ms	30 ms

7 Security Architecture

7.1 Authentication & Authorization

Table 7.1: Authentication Methods by User Type

User Type	Authentication Method & Controls
Retail Customers	JWT + password (bcrypt hashed, 10 rounds). 2FA optional (SMS/email). 8 hr timeout. 5 failed attempts → 15 min lock-out.
Corporate Clients	API keys + OAuth2 (client credentials) + IP whitelist. Key rotation 90 days. Scope-limited permissions (read/write/admin).
Store Associates	JWT + MFA (TOTP) mandatory. 12 hr timeout. Device attestation (only registered POS devices).
Field Service	JWT + biometric (fingerprint/Face ID) on mobile. Offline token (cached, 7 day expiry).
Admins	MFA (TOTP) mandatory + hardware key (YubiKey) optional. 2 hr timeout. Audit logs (all actions). Break-glass access (logged, requires 2-person approval).
Internal Services	mTLS (service mesh, Istio). SPIFFE identities. Auto cert rotation (24 hours). Mutual authentication.
Registry Auditors	OAuth2 (authorization code) + client certs. Read-only access to audit trail. Session timeout 2 hours.

7.2 RBAC Roles and Permissions

Table 7.2: RBAC Roles (7 roles)

Role	Permissions
Customer	View own orders, circular passport, circular credit balance, subscription; initiate returns; purchase products; offset carbon. No access to others' data.
Corporate Client	View corporate orders, employee roster, Scope 3 emissions reports; download ESG data; manage employee subscriptions; no access to other clients' data.
Store Associate	View store inventory, process returns (scan RFID, issue circular credit), assist with checkout, view customer orders (with consent).
Field Service	View assigned pickup tickets, update status (PENDING → PICKED _{UP} DELIVEREDREFURBISHED), <i>uploadphotos(returncondition</i>
Circular Ops	Manage take-back workflow (refurbishment grading, scrap reconciliation), update circular passport (blockchain transactions), inventory management.
Admin	User management (create/disable roles), system configuration (feature flags, rate limits), report access (all reports), audit log viewer.
Super Admin	Full system access (break-glass only, requires 2-person approval), audit log review, role assignment, DR failover authorization.

8 Conclusion

The GIVA system architecture is designed to support the company’s ambitious growth targets from \$280M (FY26E) to \$1.28B+ (FY30F) while maintaining high availability (99.95% uptime target), sub-second response times for critical operations (product search, checkout, circular passport verification), and full compliance with BIS hallmarking requirements, India CCTS, EU Digital Product Passport (DPP), and emerging circular economy regulations.

8.1 Key Architectural Decisions Summary

Table 8.1: Key Architectural Decisions & Rationale

Decision	Rationale
Microservices (vs. monolithic)	Independent scaling (product catalog, subscription billing, take-back management). Team autonomy (e-commerce, circular ops, finance). Fault isolation (billing outage doesn’t stop product catalog).
EKS (K8s) over ECS	Portability (multi-cloud for EU data sovereignty), HPA (custom metrics for demand-based scaling), larger community (CNCF, Helm, observability tools).
PostgreSQL + TimescaleDB	Single DB for transactional (customers, orders) + time-series (take-back telemetry). Reduces complexity. TimescaleDB compression (90% space savings).
Blockchain (Hyperledger Fabric) for circular passports	Immutable audit trail (regulatory for circular claims), customer trust (verifiable provenance), smart contracts (automated circular credit distribution).
React Native for mobile	Single codebase (iOS/Android) → faster development, consistent features (AR try-on, QR scan), shared logic (TypeScript). Offline support (WatermelonDB).
Hyperledger Fabric for blockchain	Permissioned (approved validators only: GIVA, BIS, Verra, Gold Standard), high throughput (1000+ TPS vs. Ethereum 15 TPS), privacy (channels for confidential data between GIVA and auditors).
Multi-region deployment	Low-latency for India (ap-south-1), US (us-east-1), Europe (eu-west-2). Data residency (GDPR, India DPDP). DR protection (region outage).

8.2 Projected Capacity by FY30F

- Active customers (monthly): 2.5M (3x growth from 850K) - Daily active users: 1,000,000 (4x growth from 250K) - Daily API requests: 10,000,000 (10x growth from 1M) - Daily orders: 50,000 (3.3x growth from 15K) - Circular take-back returns (monthly): 100,000 (3.3x growth from 30K) - Circular Passport scans (monthly): 5,000,000 (10x growth from 500K) - Blockchain transactions (daily): 100,000 (10x growth from 10K) - Data warehouse size: 50 TB (5x growth from 10 TB) - Annual revenue: \$1.28B+ (4.6x growth from \$280M) - Concurrent WebSocket connections: 25,000 (5x growth from 5K)

8.3 Architecture Readiness Assessment

The architecture is projected to handle these targets with: - **Uptime:** 99.95% (4.38 hours downtime/year), with 99.99% target for checkout and circular passport verification (52 minutes/year) - **P95 latency:** <1 sec for critical APIs (product search, checkout, circular passport verification) - **RTO:** 2 hours (DR), with 1 hour RTO for checkout and order management - **RPO:** 15 minutes (5 minutes for critical data: orders, circular credits, customer profiles) - **Scalability:** Horizontal scaling verified by load tests (10x current load)

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